

WHO IS AN INDIAN?

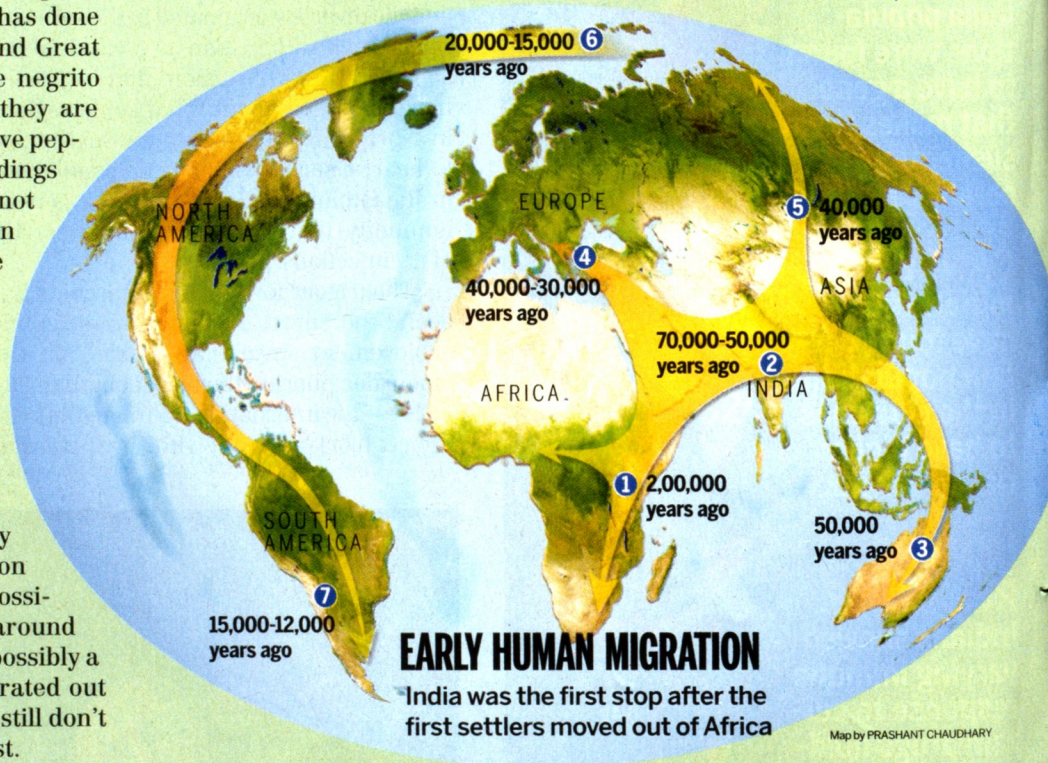
New genetic findings debunk old theories about a major Aryan influence and now talk of indigenous diversity

The Andaman and Nicobar Islands are strung out in isolated emerald splendour across the Bay of Bengal. It is their inaccessibility that has helped put them on the global genetic map. Lalji Singh, former director, Centre for Cellular and Molecular Biology (CCMB), for one believes that the islands' dwindling tribal population hold the clue to not just who were the earliest Indians but also the origin of man itself. The CCMB team has done studies among the Onge and Great Andamanese tribes whose negrito origin is evident because they are short, dark-skinned and have pepper-corn hair. Their findings showed DNA sequences did not match any other population covering either India or the rest of the world. They were unique in their origin and believed to have evolved 65,000 years ago making the Onge and Greater Andamanese tribes the earliest known humans. They do support the "Out-of-Africa" theory of prehistoric civilisation where modern humans possibly rose in East Africa around 1,50,000 years ago. After possibly a natural calamity they migrated out of Africa though scientists still don't agree where they went first.

Now a growing body of geneticists and anthropologists believe that India was a major first stop in that out-migration of humans. Earlier theories believed there were two population groups that moved out of Africa: one went further northward towards the Nile Valley and eventually to Europe and the other southwards towards India before moving out to Australia and China. Yet there is growing evi-

dence—both archaeological and genetic—to indicate that the southern wave of migration preceded that of the north. Moreover, all earliest known genetic branches are found in India and at high diversity indicating that India, after Africa, became the second great reservoir of genes. From here the genetic seeds were spread to the rest of the world. Says Partha P. Majumder, head of the Human

of Central Asia on the Indian gene pool was minor. In fact, the results of two major studies, one by CSIR's Samir Brahmachari in collaboration with major Asian countries such as China and Indonesia and the other done by Singh in collaboration with American Universities, are likely to overturn all conventional theories about India's role in the spread of the gene pool. Singh had postulated recently that the



Genetics Unit at the Indian Statistical Institute: "South Asia played a pivotal role in the out-of-Africa colonisation and dispersal of modern humans."

There is also growing evidence to indicate that the so-called Aryan invasion that brought the caste system may not have been such a major influence at all. Recent studies by Indian researchers show that their influence

"genetic diversity evolved in India itself as waves of settlers moved both north and south of the country, inter-mixing before migrating to other countries" and the results of his study to be out later this month are eagerly awaited. So after Africa, India became both a major genetic bridge and a pool for the rest of the world.

R. Pitchappan, Professor